



**UITS**  
**UNIVERSITY OF INFORMATION  
TECHNOLOGY AND SCIENCES**

## **Simulation and Modeling Lab**

**CSE-413**

### **SUBMITTED TO**

Afroja Ahmed Smrity

Lecturer of University of Information Technology and Sciences (UITS)

### **SUBMITTED BY**

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**Batch:** 53

**Semester:** Spring (2026)

**Section:** 7A2

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## Skill Mapping and Justification Form

CO–PO Mapping, Complex Engineering Problem (CEP) and Knowledge Profile (KP) Justification Report  
**Assignment/Project Title: “Designing a Simulation-Based Smart Traffic Analysis System for Urban Road Management”**

| No.        | COs                                                                         | POs   | BT             | CP/WP      | CA/EA | KP/WK      |
|------------|-----------------------------------------------------------------------------|-------|----------------|------------|-------|------------|
| <b>CO1</b> | Define various modelling terms by analyzing the given system or data.       | PO(a) | L1, L2, L4     | –          | –     | K2         |
| <b>CO2</b> | Classify various simulation models and their application in simulation.     | PO(b) | L2, L3, L4     | –          | –     | K2, K3     |
| <b>CO3</b> | Construct a model for a given set of data and examine its validity.         | PO(c) | L3, L4, L5, L6 | P1, P2, P3 | –     | K2, K3, K5 |
| <b>CO4</b> | Analyze output data produced by a model and test the validity of the model. | PO(d) | L4, L5, L6     | P1, P2, P3 | –     | K8         |

### Justification of Program Outcome (PO) through Course Outcome (COs)

| COs No.    | POs                                   | Justification                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CO1</b> | PO1 (Engineering Knowledge)           | The lab activities require students to apply fundamental concepts of mathematics, statistics, probability, and computer science to understand modeling terminology and system behavior. Students analyze system variables, entities, events, states, and performance measures while utilizing theoretical concepts necessary for representing real-world systems mathematically and computationally.                                    |
| <b>CO2</b> | PO2 (Problem Analysis)                | Students analyze different types of simulation models including deterministic, stochastic, discrete-event, continuous, and Monte Carlo models to investigate complex engineering problems. The lab requires examination of system characteristics, identification of suitable modeling approaches, and analytical comparison of alternative simulation techniques before selecting the most appropriate method for a given application. |
| <b>CO3</b> | PO3 (Design/Development of Solutions) | The lab requires students to design and construct simulation models for real-world systems based on given datasets and problem specifications. Students develop modeling structures, define system parameters, implement simulation logic, and validate model behavior to produce reliable and efficient solutions that satisfy functional requirements and support decision-making processes.                                          |
| <b>CO4</b> | PO4 (Investigation)                   | Students investigate the behavior and performance of simulated systems through output analysis and statistical validation techniques. The lab involves conducting experiments, interpreting simulation results, testing model accuracy, and evaluating the validity and reliability of models using analytical and research-oriented approaches to support engineering conclusions.                                                     |

### Justification of Complex Engineering Problem (P) and Knowledge Profile (KP)

| Attributes                    | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P1: Depth of Knowledge</b> | Solving the smart traffic analysis problem requires in-depth engineering knowledge and a first-principles approach. Students must apply theory-based engineering fundamentals (K3), including probability distributions, random number generation, statistical analysis, Monte Carlo simulation, and hypothesis testing to model uncertain traffic behavior. The assignment further requires engineering design knowledge (K5) to develop a reliable computational traffic simulation model capable of generating realistic traffic scenarios and supporting traffic management decisions. Model validation and investigation activities also involve research-oriented statistical techniques and analytical methods, thereby engaging students with research knowledge and methodologies (K8). |